

Erin Atkinson

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Education

- 2022– **PhD Candidate, Physics**, University of Toronto, Canada,
Earth, Atmospheric and Planetary physics
Submesoscale ocean fronts, filaments and vortices in multiscale flows (tentative).
- 2021–2022 **MSci Natural Sciences**, Fitzwilliam College, University of Cambridge, UK,
Specialising in Experimental and Theoretical Physics
First Class Honours
Using unsupervised classification of ocean data to track changes in Southern Hemisphere winds.
- 2018–2021 **BA Natural Sciences**, Fitzwilliam College, University of Cambridge, UK,
Specialising in Experimental and Theoretical Physics
First Class Honours
- 2011–2018 *Macmillan Academy, UK*
○ A Level - 4 A* Physics, Chemistry, Mathematics, Further Mathematics
○ GCSE - 10 A*-B Including English Language and English Literature.

Publications

Erin Atkinson and Nicolas Grisouard. “Formation and Arrest of a Surface Density Front via Strain-Driven Frontogenesis”. In: *Geophysical Research Letters* 53.12 (2026), e2025GL119220

Erin Atkinson, James Cyrus McWilliams, and Nicolas Grisouard. “Near-inertial echoes of ageostrophic instability in submesoscale filaments”. In: *Journal of Fluid Mechanics* 1015 (2025), A17

Emma JD Boland, Erin Atkinson, and Dani C Jones. “A novel heuristic method for detecting overfit in unsupervised classification of climate model data”. In: *Environmental Data Science* 2 (2023), e46

Employment and volunteer positions

- 2022– Various teaching assistant appointments
Teaching undergraduate students enrolled in physics courses offered by the Department of Physics at the University of Toronto. Primarily leading practicals and tutorials in addition to marking exam scripts and writing official solutions.
- 2024–2025 Graduate Liason Committee
Representing the interests of physics graduate students in the department in academic, supervisory and social contexts.
- 2022–2023 Doors Open Toronto, Science Rendezvous
Regular presentation at outreach events.
- 2022–2023 PGSA Internal Vice President
Executive position in the Physics Graduate Student Association at the department. Primary responsibilities include:
- Liasing with external providers to organise the daily coffee hour
 - Organising social events in the department for students, such as Halloween and Christmas parties, and the Spring Formal
- 2018–2022 Cambridge Hands-On Science
Volunteered at various outreach events, demonstrating physics experiments to members of the public.

Conferences

2026

- May Liege Colloquium on Ocean Dynamics: Submesoscale processes in the ocean
- Apr-May IHP Trimester: Mathematical Developments in Geophysical Fluid Dynamics
- Apr CIRM-IHP Joint Program Summer School: Mathematical Developments in Geophysical Fluid Dynamics
- Feb Ocean Sciences Meeting

2024

- Feb Ocean Sciences Meeting

Academic Projects

2021–2022 Masters' thesis

A project with the British Antarctic Survey. The aim was to continue previous work, automatically classifying Southern Ocean regions based on temperature data. Within the project, I demonstrated a method for utilising simulated climate data to determine the appropriate number of regions, which was previously done in an ambiguous manner.

Boland, E. J. D., Atkinson, E., & Jones, D. (2023). A Novel Heuristic Method for Detecting Overfit in Unsupervised Classification of Climate Models.

Summer Surface Science

2020 Worked with the Cavendish Laboratory Surfaces group on an online summer project. My role was to investigate reconstruction methods for under-sampled scanning helium microscope images. I learned about the physics and engineering of the Cavendish SHeM and introduced myself to machine learning, implementing a model for a Bayesian dictionary learning reconstruction method in MATLAB.